higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: SHEET METAL WORK
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 7 pages.

Copyright reserved

QUESTION 1

1.1 C
1.21 D
.3 B
1.4 D
1.5 C
1.6 A
1.7 D
1.8 D
1.9 A
1.10 C
1.11 C
1.12 D
1.13 B
1.14 B
1.15 A

(15 × 1) **[15]**

QUESTION 2

2.1 True – Because the teeth points forward

2.2 True – Because the protractor head is adjustable to any angle

2.3 False – Because when doing hacksaw work, you cut a piece of material to size and shape

2.4 False – According to the OH&S Act, workers should be supplied with PPE by the employer

2.5 True – Because sand also extinghuishes fire

2.6 False – Hot cracks occur during the cooling process

2.7 True – Because it is necessary to exhaust fumes, pollutants and moisture

2.8 False – Component numbers must correlate with the markings on the drawing

2.9 False – In a lang-lay rope, the wires and the strands are laid in the same direction

(9 × 2) **[18]**

Copyright reserved

QUESTION 3

3.1
- Haematite
- Magnetite (2)

3.2
- Carbon
- Silicon
- Sulphur
- Phosphorus
- Manganese (Any 3 × 1) (3)

3.3 3.3.1
- Crushing equipment
- Grinding mills
- Grinding balls
- Cam shafts
- Crank shafts
- Wearing plates (Any 1 × 1)

3.3.2
- Wire
- Rods
- Tubes
- Pipe fittings
- Screws
- Bolts
- Nuts (Any 1 × 1)

3.3.3
- High-tensile tubes
- High-tensile wire
- High-tensile dies for drop forgings
- High-tensile laminated springs
- High-tensile screw drivers
- High-tensile hammers
- High-tensile shear blades
- High-tensile woodwork chisels
- High-tensile connecting rods
- High-tensile locomotive tyres (Any 3 × 1)

(3 × 1) (3)

3.4 A steel wedge is inserted at the front end of the handle before the head is driven onto the handle. (2)

3.5 There are 32 teeth per 25 millimetre/32 teeth per inch on the blade. (1)

3.6
- Single cut
- Double cut
- Curved cut (3)

3.7 A pop riveter riveting two plates together. (2)

Copyright reserved

3.8 •Chuck key
•The function of the tool is to open or close a drill chuck (2)

3.9

COLOR OF CORE	PURPOSE OF CORE
Brown	Live
Blue	Neutral
Yellow/Green	Earth

(6)

3.10 The term *clearance* refers to the clearance between the top and bottom blades,✓ depending on the thickness of the material to be cut.✓ (2)

3.11 •Low-pressure system
•High-pressure system (2)

3.12 The purpose of the blowpipe is to mix the oxygen and acetylene gasses to the correct proportions. (1)

3.13 •Bend radius
•Material thickness (2)

3.14 A thermostat regulates the start-up and shut-down cycles of a heating or cooling system. (2)

3.15 85 dB (1)
[34]

QUESTION 4

4.1 •Low-carbon steel
•Medium-carbon steel
•High-carbon steel
•Tool steel (4)

4.2 Low-carbon steel:

•Girders
•Reinforcing rods
•Forgings
•Sheet metal
•Rods
•Bars
•Semi-welded tubes (Any 2 × 1)

Medium-carbon steel:

- High-tensile tubes
- High-tensile wire
- Dies
- Laminated springs
- Screwdrivers
- Hammers
- Shear blades
- Woodwork chisels
- Connecting rods
- Locomotive tyres (Any 2 × 1)

High-carbon steel:

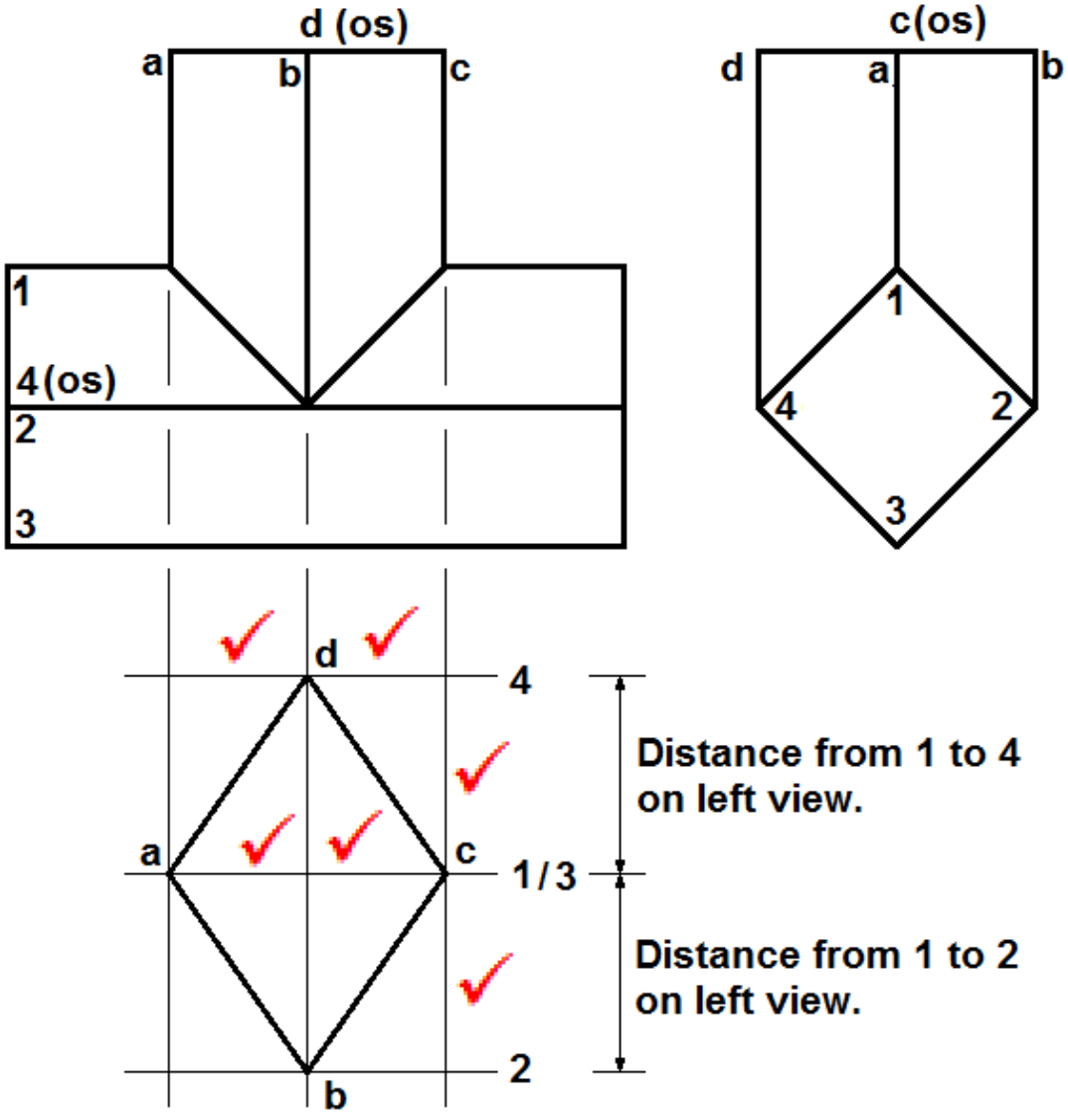
- Leaf springs
- Coil springs
- High-tensile wire ropes
- Cold chisels
- Punches
- Shear blades (Any 2 × 1)

Tool steel:

- Axes
- Picks
- Hand files
- Drills
- Cutting tools
- Ball bearings
- Razor blades (4 × 2)
  (2 × 4) (8)

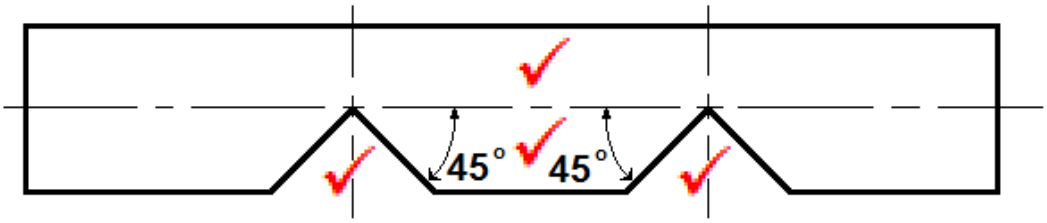
Copyright reserved

4.3



(2 Marks – Vertical spacing; 2 Marks – Horizontal spacing; 2 Marks – Shape of hole. Spacing distances to correlate with given views.) (6 × 1) (6)

4.4



(2 Marks – 2 Notches; 1 Mark – 2 Angle indications; 1 Mark – Plate) (4 × 1) (4)

4.5 Water jet cutting process (1)

4.6	Material Safety Data Sheet		(1)
4.7	The grinding process		(1)
4.8	The pressure of the gas content will increase		(1)
4.9	To prevent breathing		(1)
4.10	True		(1)
4.11	Once a month		(1)
4.12	•It can be exposed to high temperatures. •The strength of the rope is increased.		(2)
4.13	• Date of fitting • Measured diameter of the steel rope • Construction of the rope • Length of rope • Any defects on the rope • Duration of service	(Any 2 × 1)	(2) **[33]**
		TOTAL:	**100**

Copyright reserved